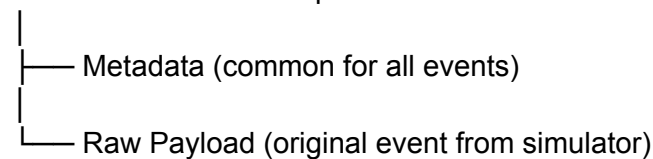

Module 1 Output

Universal Event Envelope (UEE)

Every event leaving Module 1 has **two parts**:

1. **Standard Metadata Envelope** (identical for every event)
2. **Original Payload** (unchanged, source-specific)

Universal Event Envelope



This allows Firewall, UPI, IAM, ATM, VPN, EDR, SIEM, etc. to all travel through the pipeline without losing any source-specific information.

Universal Event Envelope Structure

```
{
  "metadata": { ... },

  "raw_payload": { ... }
}
```

Part 1 — Metadata (Generated by Module 1)

These fields exist for **every single event**, regardless of whether it is Firewall, UPI, IAM or Quantum.

Field	Description
event_uuid	Internal unique UUID generated by Module 1

original_event_id	Original Event ID from source system
event_type	FIREWALL, UPI_TRANSACTION, IAM_LOGIN, EDR_ALERT, etc.
event_category	NETWORK, AUTHENTICATION, TRANSACTION, ENDPOINT, THREAT_INTEL, PAYMENT, QUANTUM
source_system	Firewall, VPN, Core Banking, SIEM, etc.
source_vendor	Cisco, Palo Alto, Fortinet, Internal Banking, Simulator, etc.
ingestion_timestamp	Time Module 1 received event
original_timestamp	Original event timestamp
processing_timestamp	Time Module 1 completed processing
schema_version	Event schema version
validation_status	VALID / INVALID
validation_errors	Missing fields if validation failed
duplicate_status	UNIQUE / DUPLICATE
duplicate_reference	Original event UUID if duplicate
processing_status	ACCEPTED / REJECTED / QUARANTINED
event_hash	SHA-256 hash for deduplication
event_size	Payload size
event_priority	LOW / MEDIUM / HIGH / CRITICAL
correlation_id	Shared incident identifier
batch_id	Batch identifier if batch ingestion
stream_id	Streaming pipeline identifier
pipeline_version	Version of ingestion pipeline

Part 2 — Entity Context

Instead of searching inside every payload later, Module 1 extracts only the **common identifiers**.

These are copied into metadata.

entity_context

Field	Nullable
customer_id	✓
employee_id	✓
username	✓
account_number	✓
card_number_masked	✓
transaction_id	✓
beneficiary_id	✓
device_id	✓
endpoint_id	✓
session_id	✓
user_id	✓

These identifiers are **not normalized**.

They are simply copied if present.

Part 3 — Network Context

Again, only copy.

network_context

Field
source_ip
destination_ip

public_ip
source_port
destination_port
protocol
country
city

These values come directly from the simulator.

Module 1 never modifies them.

Part 4 — Security Context

security_context

Field

severity
action
log_source
sensor_id
firewall_device
rule_id
signature_id

Again—

No interpretation.

Just copy.

Part 5 — Raw Payload

This is the most important part.

The entire original event is stored here **without modification**.

Example

Firewall

```
raw_payload
{
  "event_id":"FW-123",

  "source_ip":"49.37.xx.xx",

  "destination_ip":"10.0.1.12",

  "destination_port":443,

  "protocol":"HTTPS",

  "action":"ALLOW",

  "bytes_sent":2000,

  "bytes_received":1200,

  "rule_id":"FW-887"
}
```

UPI

```
raw_payload
{
  "transaction_id":"UPI-778",

  "customer_id":"C001",

  "upi_id":"abc@oksbi",

  "receiver_bank":"HDFC",

  "amount":50000,

  "merchant":"Amazon"
}
```

EDR

raw_payload

```
{
  "endpoint_id": "EP12",

  "process_name": "powershell.exe",

  "malware_name": "Emotet",

  "cpu_usage": 90,

  "memory_usage": 87
}
```

Notice that all three payloads remain completely different.

Complete Module 1 Output

UniversalEventEnvelope

|

|— metadata

|

- |— event_uuid
- |— original_event_id
- |— event_type
- |— event_category
- |— source_system
- |— source_vendor
- |— ingestion_timestamp
- |— original_timestamp
- |— processing_timestamp
- |— validation_status
- |— duplicate_status
- |— schema_version
- |— event_hash
- |— correlation_id
- |— pipeline_version
- |— processing_status

|

└─ entity_context

└─ customer_id
└─ employee_id
└─ account_number
└─ transaction_id
└─ session_id
└─ device_id
└─ endpoint_id
└─ beneficiary_id

|

└─ network_context

└─ source_ip
└─ destination_ip
└─ source_port
└─ destination_port
└─ protocol
└─ country
└─ city

|

└─ security_context

└─ severity
└─ action
└─ rule_id
└─ sensor_id
└─ signature_id
└─ log_source

|

└─ raw_payload

└─ Original event (unchanged)